Temporal Interitem Relations in Visual Working Memory

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Introduction

- ✓ Temporal information of sequentially presented items is important for memory formation and retention (Lewandowsky & Murdock, 1989)
- ✓ The temporal order of objects and the temporal intervals associated with objects can be automatically learned and stored
- ✓ Ordinal and relational transformation in spatiotemporal structure impairs visual working memory performance

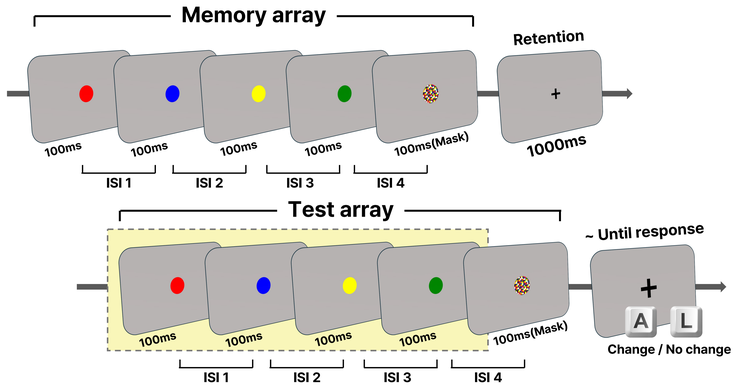
Question

When items are sequentially presented, how does the relative information of the temporal structure impact visual working memory?

Method

✓ **Change detection task**

: report whether four colors in the test array match the memory array, regardless of the order of items



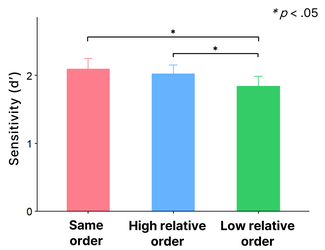
Exp 1 : Item order

(N = 30)

Memory array		Absolute order change	Relational order change
● – ● – ● – ●	A B C D	0	0

Item order	Test array			
Same order	● – ● – ● – ●	A B C D	0	0
High relative order	● – ● – ● – ●	B C D A	6 (1+1+1+3)	4 (1+0+1+2)
	● – ● – ● – ●	C D A B	8 (2+2+2+2)	4 (1+1+1+1)
	● – ● – ● – ●	D A B C	6 (3+1+1+1)	4 (2+1+0+1)
Low relative order	● – ● – ● – ●	B D A C	6 (+1+2+2+1)	8 (2+2+2+2)
	● – ● – ● – ●	C A D B	6 (2+1+1+2)	8 (2+2+2+2)
	● – ● – ● – ●	D C B A	8 (3+1+1+3)	8 (2+2+2+2)

✓ **ISI 1, 2, 3, 4 : 100ms**



✓ Temporal representation of items in WM relies on the relative order information between items

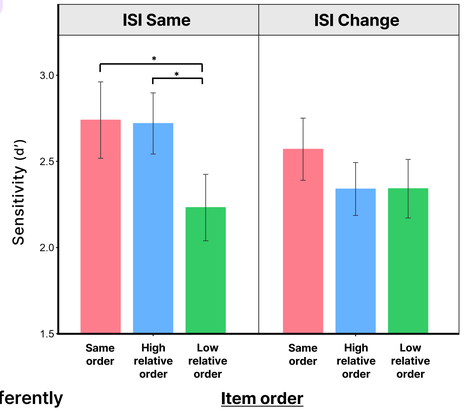
Exp 2 : Item order x ISI order

(N = 33)

✓ **ISI 1, 2, 3 : 100 & 300 & 600ms**

Memory array	Absolute order change	Relational order change
ISI 1 – ISI 2 – ISI 3	0	0

ISI order			
ISI same	ISI 1 – ISI 2 – ISI 3	0	0
ISI change	ISI 2 – ISI 3 – ISI 1	4 (1+1+2)	4 (1+1+2)
	ISI 3 – ISI 1 – ISI 2	4 (2+1+1)	4 (2+1+1)
	ISI 1 – ISI 3 – ISI 2	2 (0+1+1)	5 (1+2+2)
	ISI 2 – ISI 1 – ISI 3	2 (1+1+0)	5 (2+2+1)



✓ Relative order between items affects WM differently depending on the temporal interval change

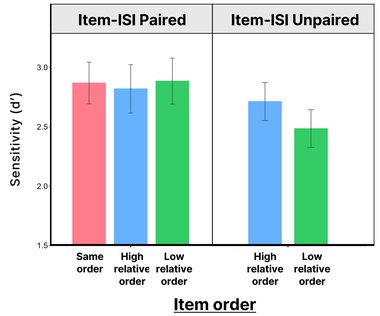
Exp 3 : Relative order x Item-ISI pairing

(N = 32)

✓ **ISI 1, 2, 3, 4 : 100 & 200 & 400 & 800ms**

Memory array: ● — ● — ● — ● — | A 1 | B 2 | C 3 | D 4

Test array	Item-ISI paired	Item-ISI unpaired
High relative order	B 2 C 3 D 4 A 1	B 4 C 1 D 2 A 3
	D 4 A 1 B 2 C 3	D 2 A 3 B 4 C 1
Low relative order	B 2 D 4 A 1 C 3	B 3 D 1 A 4 C 2
	C 3 A 1 D 4 B 2	C 2 A 4 D 1 B 3



✓ Items and subsequent intervals were stored as associations in WM, enhancing performance

Discussion

- ✓ Temporal representation of items in WM could be constructed by relative temporal order with the associated temporal interval
- ✓ Temporal interval information may serve as an identity that facilitates detection performance

References

- Lewandowsky, S., & Murdock Jr, B. B. (1989). Memory for serial order. *Psychological Review, 96*(1), 25.
- Heuer, A., & Rolfs, M. (2023). Temporal and spatial reference frames in visual working memory are defined by ordinal and relational properties. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 49*(9), 1361.
- van de Ven, V., Jäckels, M., & De Weerd, P. (2022). Time changes: Timing contexts support event segmentation in associative memory. Psychonomic Bulletin & Review, 1-13.

Acknowledgments This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2021S1A5A2A01062955)

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